Spatial Query Translation

MapMyIndia
Technical Architecture Proposal

February 2026
Executive Presentation

Problem Analysis

Problem

Current map systems suffer from catastrophic spatial prioritization failures

Local Intent Failure

User Query

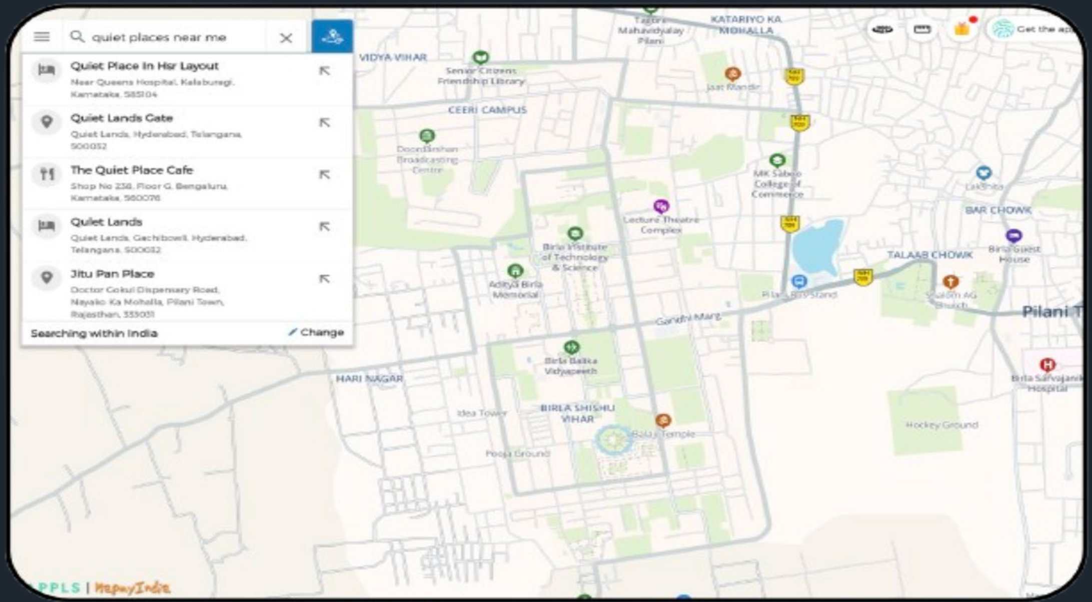
"quiet places near me" in Pilani

↓

System Returns

Results from Bengaluru, Hyderabad

1,000+ km away



Exact Keyword Match

System matches "Quiet Lands Gate" instead of understanding "quiet" as a characteristic (low traffic, low noise, calm environment)

Trust Erosion

If core "near me" functionality fails, users abandon platform for competitors.

This is not a UI problem. It is an architecture problem.

Current engine: Deterministic keyword filtering, not intent modeling. The system requires fundamental architectural transformation to handle natural language spatial queries effectively.

Solution Architecture • Option 1

Frontier Model Per Request- Grok

GPT-class model with dynamic SQL generation

2-3
Weeks Development
Rapid prototyping

$$$
Marginal Cost
Per-request inference

5-10%
Hallucination Risk
SQL errors

Pros

- **Extremely easy to develop**
  Minimal engineering effort required
- **Fast to prototype**
  Quick proof-of-concept validation
- **No schema engineering**
  LLM handles structure dynamically

Cons

- **Every request incurs cost**
  Unsustainable
- **LLMs hallucinate**
  Columns, table names, filters wrong
- **Hard to guarantee determinism**
  Latency spikes under load

Verdict: This approach does not scale economically or architecturally. Inference costs at scale become prohibitive, and hallucination risks compromise data integrity.

Solution Architecture • Option 2

Custom In-House Spatial LLM

Fully trained proprietary model for spatial intent + SQL reasoning

6-12
Months Development
Full model training + infra build

$$$
Infrastructure cost
Capital Intensive Training + GPU clusters

HIGH
Execution Risk
Model underperformance + data dependency

Pros

- **Full IP ownership**
  End-to-end proprietary model
- **Maximum architectural control**
  Optimized specifically for spatial intent
- **Long-term cost efficiency**
  No external inference dependency

Cons

- **Long time to develop**
  Uncharted territory
- **High upfront capital GPU infra**
  Expensive and hardly available
- **Data dependency risk**
  Requires large labeled spatial dataset

Verdict: High strategic value but capital heavy and slow to develop. Not suitable for rapid execution

Solution Architecture • Option 3 (Recommended)

Local RAG Optimized Intent Model

Open-source LLM fine tuned for spatial intent extraction

$0
Marginal Cost
Local inference

30-50%
Lower Latency
No remote API calls

100%
Deterministic
SQL generation

Pros

- **Runs locally** — No per-request external API cost
- **Structured JSON output** — Reduces hallucination risk
- **Deterministic validation** — Prevents SQL errors
- **Full architectural control** — Defensible and scalable

Cons

- **Higher initial complexity** — More engineering effort upfront
- **Requires schema indexing** — RAG infrastructure needed
- **Intent validation layer** — Additional component to build

Verdict: This approach is scalable, controllable, and defensible. Higher initial investment pays off through zero marginal costs, lower latency, and deterministic behavior at scale.

Our Approach

Local RAG Intent Engine Architecture

Restricting probabilistic boundary to intent extraction only

Architecture Flow

1. **Natural Language Query**
   "quiet cafes within 2km"

↓

2. **RAG + Local LLM**
   Intent extraction only → Strict JSON

↓

3. **Deterministic Validation**
   Schema check → Template selection

↓

4. **Spatial Execution**
   PostGIS → Parameter binding → Results

Benchmark Results

Incorrect Spatial Matches -40%

Latency Reduction -35%

✓ 0 SQL Hallucinations

✓ 0 Unsafe Queries

Key Principles

- AI only for intent extraction
- Everything else deterministic
- Linear economic scaling

JSON Output Example

{type: "search", target_type: "cafe", filters: {quiet: true}, radius: 2000, ...}

No Free Text

No SQL Generation

User Experience

Conversational Discovery Interface

Transforming search from keyword matching to intelligent intent understanding

Conversational Search Bar

quiet place to study near me

best chai within 5km

less crowded cafes

Intent Preview

Searching for:

quiet | cafes | within 2km | study-friendly

Confidence Indicator

92% High Confidence

Real-time scoring from intent extraction model

Smart Nearby Ranking

Results ranked by multiple signals:

1. **Intent Relevance**
   Semantic match to query
2. **Co-occurrence Frequency**
   Behavioral patterns
3. **Experiential Attributes**
   Quiet, crowd level, ambiance
4. **Spatial Proximity**
   Distance from user

Transformation: MapMyIndia shifts from keyword-based search to intelligent discovery, understanding not just what users type, but what they actually mean.

Work so far

Built

Transforming search from keyword matching to intelligent intent understanding

Conversational Search Bar

quiet place to study near me

best chai after 10pm

less crowded temples today

Confidence Indicator

92% High Confidence

Real-time scoring from intent extraction model

Intent Preview

Searching for:

quiet cafes within 2km study-friendly

Smart Nearby Ranking

Results ranked by multiple signals:

1. **Intent Relevance**
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Implementation Roadmap

Foundation Phase: Weeks-14

Building schema grounding and intent extraction infrastructure

W1-2 Schema Grounding + RAG

Infrastructure foundation

- **Build schema_docs index**
  Extract database schema, column names, enums, relationships
- **Vector embedding pipeline**
  Tables, enums, constraints → Vector database
- **RAG retrieval pipeline live**
  vector_search(raw_query → schema_docs)

Goal: LLM always grounded in schema context with allowed intents, valid filters, parameter constraints

W3-4 Intent Extraction Engine

Core AI component

- **Fine-tune open-source LLM**
  Optimized for spatial intent understanding
- **Strict JSON output enforced**
  No free text, no SQL generation
- **Retry logic implemented**
  Re-extraction if JSON invalid

Output Schema:

{type, reference, target_type, filters, radius, order, limit}

Schema Context
Always grounded

Probabilistic Only
Intent extraction

Structured Output
Zero SQL

Implementation Roadmap

Core Development: Weeks 5–10

Validation, spatial execution, and research-backed ranking

W5-6 Validation + Templates

Intent Validator
- Required fields check
- Enum correctness
- Numeric bounds
- Injection safety
- Entity existence

SQL Template Registry
intent.type → fixed template

Parameter Binding
Prepared statements only

Result: Determinism boundary established

W7-8 Spatial Execution

Postgres + PostGIS
Spatial database integration

GiST Spatial Indexing
Efficient nearest-neighbor search

Spatial Functions
- ST_Distance
- ST_Buffer
- ST_Within

Performance Target
O(n log n) spatial search

W9-10 Top-K Algorithm

Research-Backed
"Efficient processing of top-k frequent spatial keyword queries"

Scientific Reports, Nature

Keyword Co-occurrence
Learn frequent keyword combinations

Spatial Attributes
Which attributes appear together

Ranking Improvement
Frequency + spatial proximity

Impact: Intent relevance, result ranking, multi-word understanding

Zero SQL Errors

O(n log n) Search

Research-Backed

Smarter Ranking

Implementation Roadmap

Integration & Deployment: Weeks 11-12

Production deployment and end-to-end validation

W11-12

API Gateway + Frontend

Production deployment

API Gateway Live

Centralized request routing and rate limiting

Intent Service Deployed

LLM + RAG pipeline production-ready

UI Integration Complete

Conversational search, intent preview, confidence scores

End-to-End Flow: text → intent(JSON) → template(SQL) → PostGIS → rows → JSON

API Gateway

Intent Service

Validator

Query Planner

Spatial DB

Response

Strategic Positioning

Building the Future of Map Intelligence

Our Architecture

- **Deterministic Core + AI Edge**
  AI restricted to intent extraction only
- **Zero Hallucinated SQL**
  100% deterministic query generation
- **Economically Scalable**
  Zero marginal inference cost
- **Research-Backed Ranking**
  Peer-reviewed spatial keyword algorithms

Transformation

MapMyIndia shifts from **keyword search** to **intelligent spatial reasoning**. We don't just match words—we understand intent, context, and experiential needs.

Our Unique Advantage

Schema-Grounded Intent Engine
RAG-optimized with deterministic boundaries

Deterministic Spatial Planning
Validated SQL templates with zero hallucination

Research-Backed Ranking Model
Top-k frequent spatial keyword algorithms

Cost-Efficient Local Inference
Zero marginal cost at scale